Exercise 3: Clustering Using PCA and Vectorization Techniques

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Context

You are tasked with analyzing two datasets:

1. **BBC News dataset**: Contains text data grouped into 5 categories: **Business**, **Entertainment**, **Politics**, **Sport**, and **Tech**.
2. **20NewsGroups dataset**: Contains text data grouped into 20 topics.

The goal is to transform the data into word embeddings, apply clustering techniques, and evaluate the results. Improve the results by using **PCA** for dimensionality reduction.

Problem Statement

Cluster the articles in **both datasets** using the following methods:

1. **Feature extraction**: Use **TF-IDF**, **FastText**, and **Word2Vec** to create embeddings.
2. **Clustering algorithms**: Implement **K-Means**, **HDBSCAN**, and **Agglomerative Clustering**.
3. **Evaluation metrics**: Evaluate the clustering results using **NMI (Normalized Mutual Information)**, **ARI (Adjusted Rand Index)**, **AMI (Adjusted Mutual Information)**, and the **Silhouette Score** or **Elbow Method**.

Tasks

1. **Transform the Data**:
    - For each dataset (BBC News and 20NewsGroups), transform the text data using **TF-IDF**, **FastText**, or **Word2Vec**.
    - You can use one or all the methods and compare them to determine which works best.
    - **Note**: No preprocessing of text is required in this exercise.
2. **Apply Clustering Algorithms**:
    - Implement **K-Means**, **HDBSCAN**, and **Agglomerative Clustering** for each dataset.
    - Apply these algorithms to the embeddings obtained from the feature extraction methods.

3. **Evaluate and Interpret Results**:
   - Compare clustering results using **NMI**, **ARI**, and **AMI** metrics.
   - Interpret the performance of each algorithm with each embedding method.
4. **Fine-Tune with PCA**:
   - Use PCA to reduce the dimensionality of the embeddings.
   - Find the optimal reduction to achieve the best clustering results (i.e., clusters near the ground truth numbers: 5 for BBC News and 20 for 20NewsGroups).
   - **Bonus:** Visualize the clusters in 2D space for better interpretability using t-SNE.
5. **Visualization**:
   - Store the results of your clustering experiments in a dictionary or DataFrame.
   - Create bar plots or similar visualizations to compare the performance metrics (e.g., NMI, ARI, AMI) across the different methods and datasets. This step is for training yourself in plots

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Deliverables

1. **Jupyter Notebook**:
   - Include documented code for feature extraction, clustering, evaluation, and visualization.
   - Provide commentary on the comparison of techniques and performance metrics.
2. Ensure the notebook contains:
   - **Markdown explanations** for each step.
   - Well-organized and easy-to-follow code.

Hints and Notes

1. **Datasets**:
   - The **BBC News dataset** has 5 categories; you expect to find **5 clusters** as ground truth. Use the **bbc_news_test.csv** to load the dataset

- The **20NewsGroups dataset** has 20 topics; you expect to find **20 clusters** as ground truth. Use the snippet provided in file: **load_20newsgroups.py** in order to load the dataset

2. **Feature Extraction**:
    - You can choose to use only one embedding method (e.g., TF-IDF, Word2Vec, or FastText), but you must **justify your choice**.
3. **Clustering Results**:
    - Do not expect perfect clustering with the number of clusters matching the ground truth, but aim for **high evaluation metric scores** (e.g., AMI, NMI, ARI).
